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Hour-Minute-Second Timer

Digital Design and Verification

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Hour-Minute-Second Timer

Overview

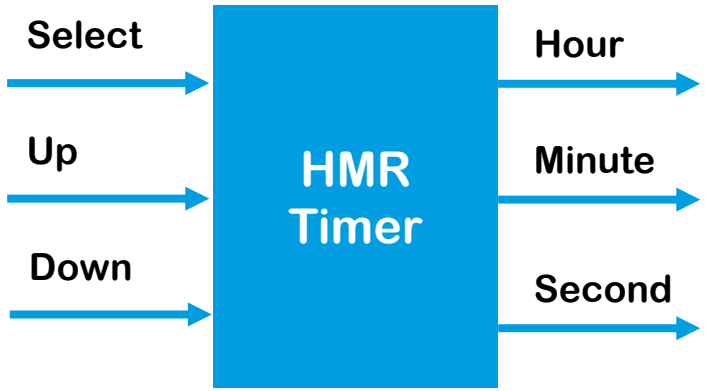
- Input
  - Functional specification (this slide)
- Output
  - Detail Design document
  - RTL source code (Synthesizable Design)
  - [Optional] Verification (Environment, Result)

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Hour-Minute-Second Timer

Requirement & Functional Specification

- Design a simple Hour-Minute-Second Timer
  - Output the real time value via Hour, Minute, Second output signals
  - Time value can be adjusted by using Select, Up, Down input signals
    - Select: select to adjust hour, minute, second value
    - Up: Increase time value
    - Down: Decrease time value
  - Select, Up and Down are edge-triggered signals
  - 1Mhz input clock and asynchronous active low reset

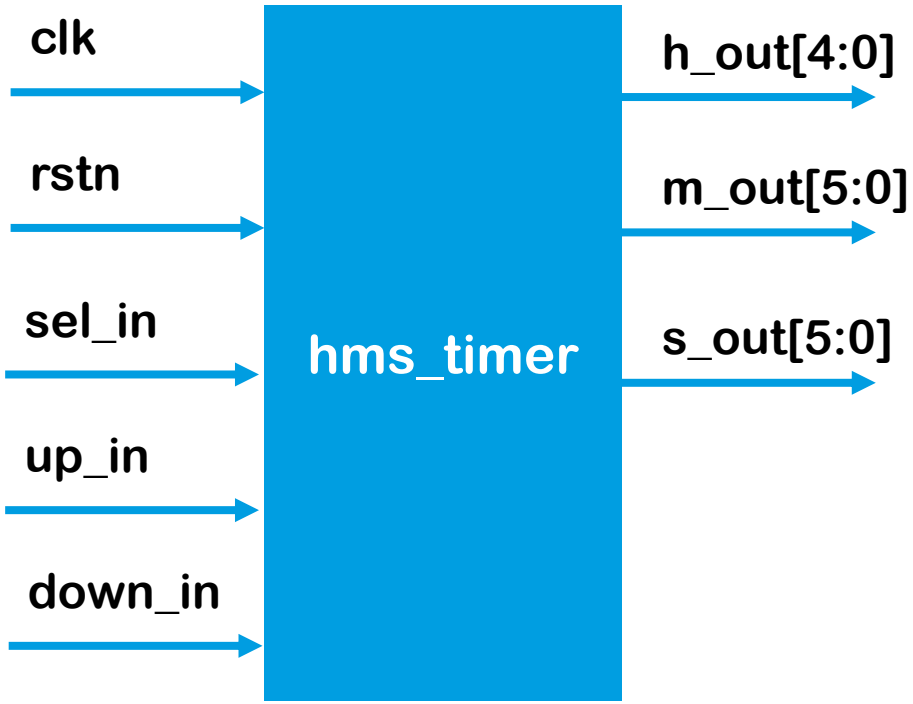


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Hour-Minute-Second Timer

I/O

Port Name	I/O	Width	Description
clk	In	1	Clock signal
rstn	In	1	Reset signal (Asynchronous active low)
sel_in	In	1	hour/minute/second Select signal
up_in	In	1	Time value increase trigger
down_in	In	1	Time value decrease trigger
h_out	Out	5	Output hour value
m_out	Out	6	Output minute value
s_out	Out	6	Output second value



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